|INCH-POUND|

MIL-M-28787/308A
AMENDMENT 3
30 July 1993
SUPERSEDING
AMENDMENT 2
11 FEBRUARY 1986

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

PROGRAM CONTROL UNIT, KEY CODE MJC

This amendment forms a part of Military Specification MIL-M-28787/308A(EC), dated 23 July 1981, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

4.4.2: Delete in its entirety and substitute:

"4.4.2 Order of testing. Test conditions A through LL shall be performed in any order with the following exceptions. Test conditions A, B, C, D, and E are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, and E where applicable, followed by the remaining tests in the sequence."

PAGES 4, 5, 6, 7, AND 8

*TABLE I: Delete and substitute new table I as printed on pages 4, 5, 6, 7, and 8 of this amendment.

PAGE 9

*4.5.1, fifth and sixth lines: Delete "D through AA and DD through LL" and substitute "F through AA and DD through LL".

*4.5.3, third line: Delete "D through AA" and substitute "F through AA".

*4.5.3, sixth line: Delete "D through LL" and substitute "F through LL".

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PAGE 24

FIGURE 9: Delete and substitute new figure 9 as printed on page 9 of this amendment.

PAGE 32

FIGURE 12: Delete and substitute new figure 12 as printed on page 10 of this amendment.

PAGE 34

*TABLE XI, Test conditions K1-K4, Min limit: Delete "-1" and substitute "-50".

*TABLE XII, Test conditions L1-L4, Max limit: Delete "+1" and substitute "+50".

PAGE 40

*FIGURE 17: Delete and substitute new figure 17 as printed on page 11 of this amendment.

PAGE 72

FIGURE 31, NOTES: Add new note as follows: "3. Setups are performed by first connecting points A and B, then momentarily connecting point C to pin 4."

PAGE 81

FIGURE 33, NOTES: Delete note 3 in its entirety.

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Custodians:
Army - ER
Navy - NW
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0280)

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TABLE I Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/ - 3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation Initial End-of-life	II II	S S	10 1	INF INF	10 1	INF INF	10 1	INF INF	10 1	INF INF	10 1	INF INF	MΩ MΩ
B	Voltage (VCC) stress	III	Q	<----------Must pass all remaining tests---------->										
C1-C16	Input breakdown current (IIB)	IV	S	0 0	1.0	0 0	1.0	0 0	1.0	0.0	1.0	0.0	1 0	mA
D1-D16	Input clamp diode voltage	V	S	-0 4	-1 5	-0 4	-1.5	-0.4	-1.5	-0.4	-1 5	-0.4	-1.5	V
E1-E10	Output short circuit current (IOS)	VI	S	-30	-85	-30	-85	-30	-85	-30	-85	-30	-85	mA
F1-F4	Input current, high level	VII	X	0	20	0	20	0	20	0	20	0	20	µA
F5 F10,F12, F14,F15,		VII	X	0	40	0	40	0	40	0	40	0	40	µA
F11		VII	X	0	90	0	90	0	90	0	90	0	90	µA
F13		VII	X	0	50	0	50	0	50	0	50	0	50	µA
F16	Input current, high level	VII	X	0	250	0	250	0	250	0	250	0	250	µA
G1-G4	Input current, low level	VIII	X	0 000	-0 398	0 000	-0.398	0.000	-0.398	0.000	-0.398	0.000	-0.398	mA
G5-G10,G12, G13,G15		VIII	X	0 000	-0 775	0.000	-0.775	0 000	-0 775	0.000	-0.775	0.000	-0.775	mA
G11		VIII	X	0 00	-2 55	0 00	-2 55	0.00	-2.55	0.00	-2.55	0.00	-2 55	mA
G14		VIII	X	0 000	-0 725	0 000	-0 725	0 000	-0 725	0.000	-0.725	0.000	-0 725	mA
G16	Input current, low level	VIII	X	0 00	-3.59	0 00	-3 59	0.00	-3.59	0.00	-3.59	0.00	-3 59	mA
H1 H10	Output voltage, high level	IX	X	2.4	4 5	2 4	4.5	2.4	4.5	2.4	4.5	2.4	4 5	V
J1-J10	Output voltage, low level	X	X	0.0	0.5	0.0	0 5	0.0	0.5	0.0	0 5	0.0	0.5	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ - 3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°		-55°C ± 5°C		85°C ± 5°		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
K1-K4	High impedance leakage current, high level	XI	X	-50	+50	-50	+50	-50	+50	-50	+50	-50	+50	µA
L1 L4	High impedance leakage current, low level	XII	X	-50	+50	-50	+50	-50	+50	-50	+50	-50	+50	µA
M	Power supply current	XIII	X	10	208	10	218	10	218	10	238	10	231	mA
N1-N6	Functional tests	XIV	X	<----------See table XIV and notes---------->										
P1-P10	Transition time, tTLH	XVI	S	2	17	2	17	2	21	2	17	2	25	ns
Q1 Q10	Transition time, tTHL	XVII	S	1	18	1	18	1	21	1	18	1	24	ns
R1	Propagation delay time, tPLH	XVIII	S	7	70	7	70	7	79	7	70	8	88	ns
R2,R3		XVIII	S	5	58	5	58	6	66	5	59	7	74	ns
R4		XVIII	S	6	65	6	65	7	73	6	65	8	82	ns
R5		XVIII	S	7	78	7	78	8	87	7	78	9	97	ns
R6		XVIII	S	6	69	6	69	7	78	6	69	8	87	ns
R7		XVIII	S	5	53	5	53	6	60	5	53	6	68	ns
R8,R11		XVIII	S	3	39	3	39	4	45	3	39	5	52	ns
R9		XVIII	S	4	46	4	46	5	53	4	46	6	60	ns
R10		XVIII	S	6	61	6	61	6	69	6	61	7	78	ns
R12		XVIII	S	2	29	2	29	3	33	2	29	3	37	ns
R13		XVIII	S	2	21	2	21	2	25	2	21	3	30	ns
R14		XVIII	S	3	36	3	36	4	41	3	36	4	46	ns
R15,R16	Propagation delay time, tPLH	XVIII	S	2	17	2	17	2	20	2	17	2	23	ns

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ - 3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R17	Propagation delay time, tPLH	XVIII	S	5	59	5	59	6	66	5	59	7	74	ns
R18,R19		XVIII	S	4	46	4	46	5	52	4	46	5	58	ns
R20		XVIII	S	5	53	5	53	5	59	5	53	6	66	ns
R21		XVIII	S	6	66	6	66	7	75	6	66	8	84	ns
R22		XVIII	S	3	38	3	38	4	43	3	38	4	48	ns
R23,R24		XVIII	S	4	47	4	47	5	53	4	47	6	60	ns
R25		XVIII	S	4	43	4	43	4	49	4	43	5	55	ns
R26,R29		XVIII	S	2	28	2	28	3	33	2	28	3	38	ns
R27		XVIII	S	3	35	3	35	4	40	3	35	4	45	ns
R28	Propagation delay time, tPLH	XVIII	S	5	50	5	50	5	57	5	50	6	65	ns
S1	Propagation delay time, tPHL	XIX	S	7	70	7	70	7	79	7	70	8	88	ns
S2,S3		XIX	S	5	58	5	58	6	66	5	58	7	74	ns
S4		XIX	S	6	65	5	65	7	73	6	65	8	82	ns
S5		XIX	S	7	78	7	78	8	87	7	78	9	97	ns
S6		XIX	S	6	69	6	69	7	78	6	69	8	87	ns
S7		XIX	S	5	53	5	53	6	60	5	53	6	68	ns
S8,S11		XIX	S	3	39	3	39	4	45	3	39	5	52	ns
S9		XIX	S	4	46	4	46	5	53	4	46	6	60	ns
S10	Propagation delay time, tPHL	XIX	S	6	61	6	61	6	69	6	61	7	78	ns

- See footnotes at end of table.

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TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/ - 3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		80°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S12	Propagation delay time, tPHL	XIX	S	2	29	2	29	3	33	2	29	3	37	ns
S13		XIX	S	2	21	2	21	2	25	2	21	3	30	ns
S14		XIX	S	3	36	3	36	4	41	3	36	4	46	ns
S15,S16		XIX	S	2	17	2	17	2	20	2	17	2	23	ns
S17		XIX	S	5	59	5	59	6	66	5	59	7	74	ns
S18,S19		XIX	S	4	46	4	46	5	52	4	46	5	58	ns
S20		XIX	S	5	53	5	53	5	59	5	53	6	66	ns
S21		XIX	S	6	66	6	66	7	75	6	66	8	84	ns
S22		XIX	S	3	38	3	38	4	43	3	38	4	48	ns
S23,S24		XIX	S	4	47	4	47	5	53	4	47	6	60	ns
S25		XIX	S	4	43	4	43	4	49	4	43	5	55	ns
S26,S29		XIX	S	2	28	2	25	3	33	2	28	3	38	ns
S27		XIX	S	3	35	3	35	4	40	3	35	4	45	ns
S28	Propagation delay time, tPHL	XIX	S	5	50	5	50	5	57	5	50	6	65	ns
T1-T4	Disable recovery time, tPLZ	XX	S	5	55	5	55	6	60	5	55	6	65	ns
U1-U4	Enable recovery time, tPZL	XXI	S	5	50	5	50	5	55	5	50	6	60	ns
V1-V4	Disable recovery time, tPHZ	XXII	S	5	55	5	55	6	60	5	55	6	65	ns
W1-W4	Enable recovery time, tPZH	XXIII	S	5	50	5	50	5	55	5	50	6	60	ns

- See footnotes at end of table.

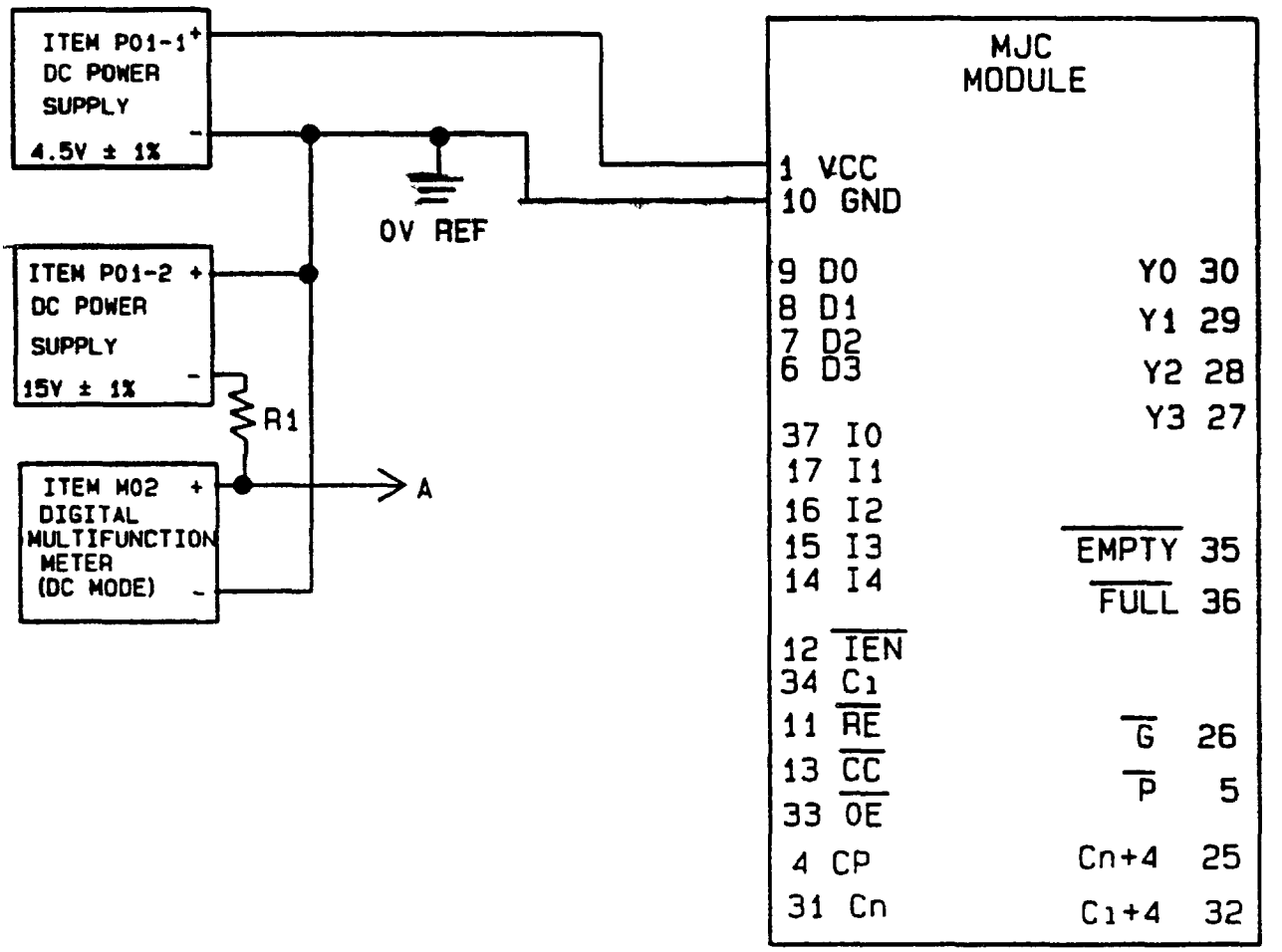
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Test condition	Parameter	Ref test table	Test type 1/ - 3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		80°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
X1-X9	Output crosstalk, low level	XXIV	Q	0	500	0	500	0	500	0	500	0	500	mV
Y1-Y8	Output crosstalk, high level	XXV	Q	0	500	0	500	0	500	0	500	0	500	mV
Z	Power supply transient amplitude	XXVI	Q	0	400	0	400	0	400	0	400	0	400	mV
AA	Power supply filter capacitance	XXVII	S	3.15	5.85	3.15	5.85	3.15	5.85	3.15	5.85	3.15	5.85	µF
BB1-BB4	Input capacitance	XXVIII	Q	6	55	6	55	6	55	6	55	6	55	pF
BB5-BB11, BB13-BB16		XXVIII	Q	2	25	2	25	2	25	2	25	2	25	pF
BB12	Input capacitance	XXVIII	Q	4	40	4	40	4	40	4	40	4	40	pF
CC1 CC10	Output capacitance	XXIX	Q	2	30	2	30	2	30	2	30	2	30	pF
DD1-DD4	Enable recovery time, tPZL	XXX	S	8	80	8	80	10	100	8	80	11	108	ns
EE1-EE4	Disable recovery time, tPLZ	XXXI	S	6	60	6	60	7	70	6	60	8	76	ns
FF1-FF4	Disable recovery time, tPHZ	XXXII	S	6	60	6	60	7	70	6	60	8	76	ns
GG1-GG4	Enable recovery time, tPZH	XXXIII	S	8	80	8	80	10	100	8	80	11	108	ns
HH1-HH4	Enable recovery time, tPZL	XXXIV	S	6	60	6	60	7	70	6	60	8	76	ns
JJ1-JJ4	Disable recovery time, tPLZ	XXXV	S	5	50	5	50	6	60	5	50	7	66	ns
KK1 KK4	Enable recovery time, tPZH	XXXVI	S	6	60	6	60	7	70	6	60	8	76	ns
LL1-LL4	Disable recovery time, tPHZ	XXXVII	S	5	50	5	50	6	60	5	50	7	66	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

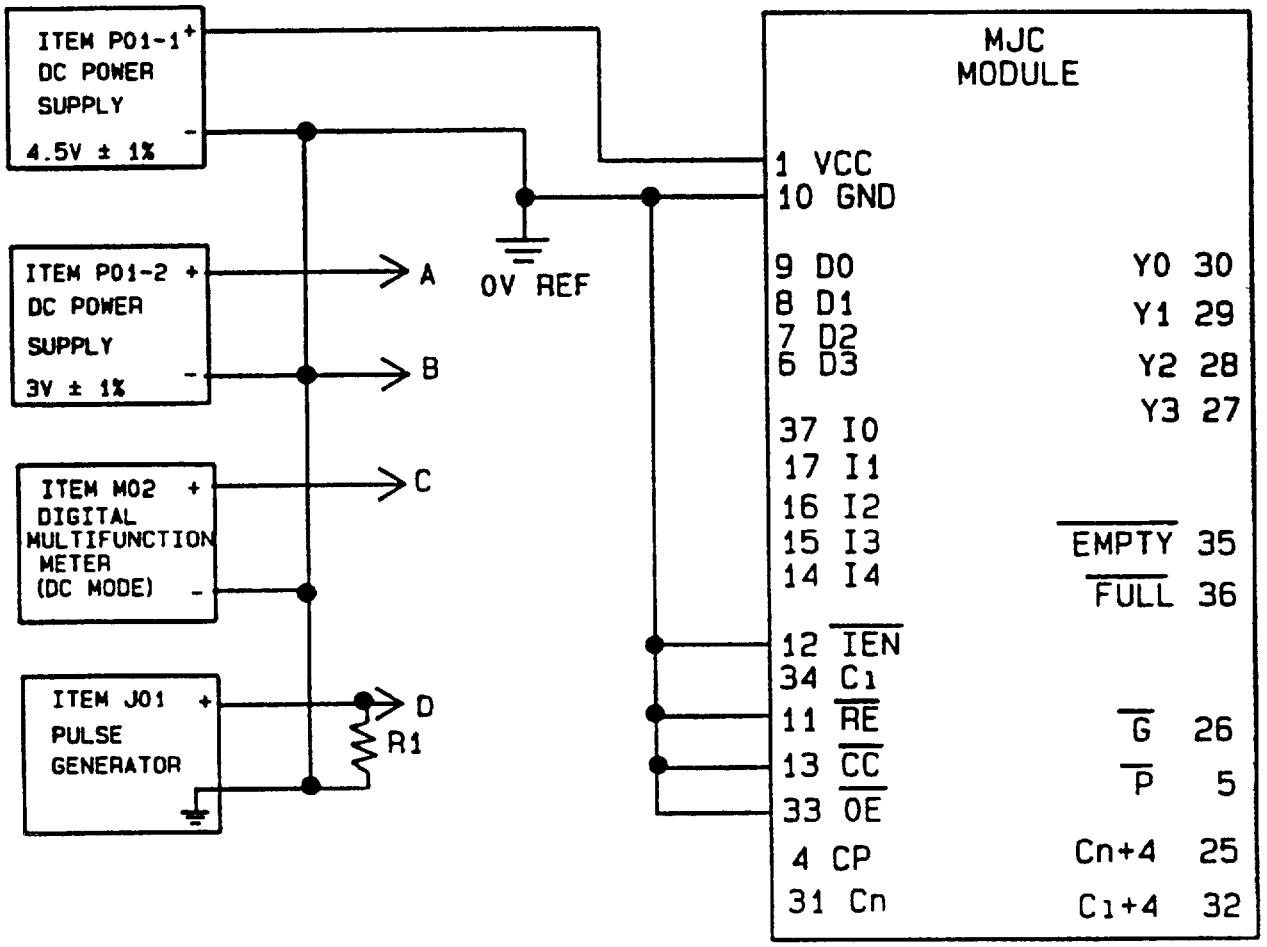
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NOTE: R1 = 750Ω ± 1%, 1/4 W.

FIGURE 9. Input clamp diode voltage test circuit for table V.

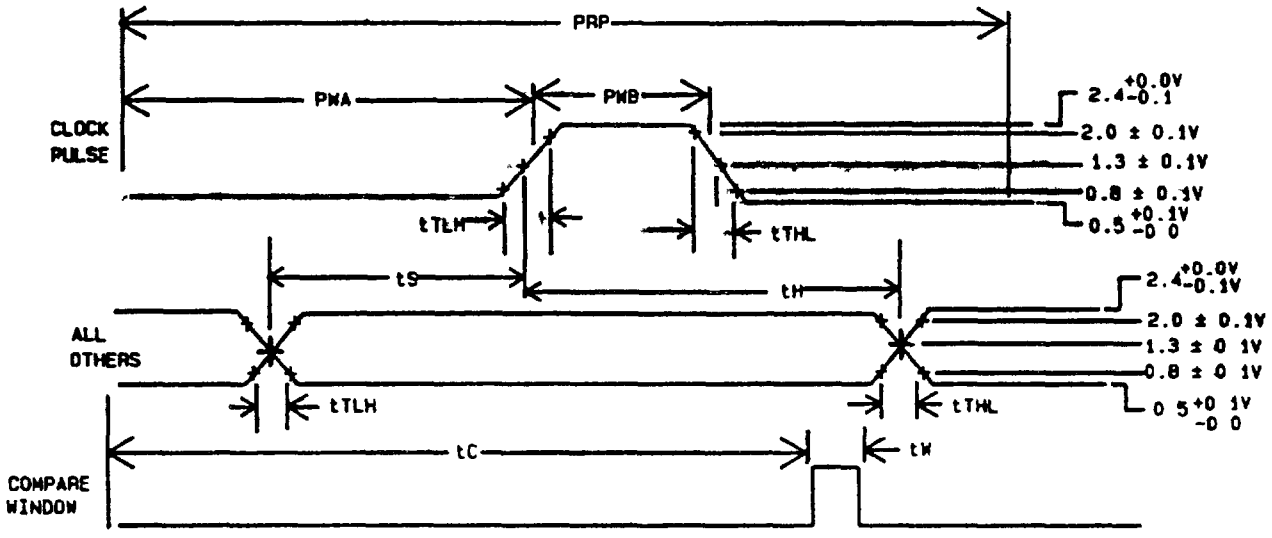
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NOTES:
1. R1 = 51Ω ± 5%, 1 W.
2. Load all outputs per figure 13.
3. Item J01 is used in the single pulse mode. When point D is referenced, apply a single pulse per figure 3.
4. For test condition J9, apply 17 single pulses after point D is referenced.

FIGURE 12. Output voltage (high-level and low-level) test circuit for tables IX and X.

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NOTE: Pulse parameters:

PRP = 376 ± 2 ns.
PWA = 95 ± 2 ns.
PWB = 36 ± 2 ns

tC = 350 ± 2 ns ($\overline{\text{FULL}}$).
= 230 ± 2 ns (All others).
tW = 10 ± 2 ns.
tTLH = tTHL = 5 ± 5 ns.

tH = 5 ± 1 ns ($\overline{\text{RE}}$).
= 120 ± 2 ns (All others).

tS = 125 ± 1 ns (I0-I4).
= 81 ± 1 ns ($\overline{\text{CC}}$).
= 70 ± 1 ns ($\overline{\text{IEN}}$).
= 53 ± 1 ns (Cn).
= 38 ± 1 ns (Ci).
= 74 ± 1 ns (D0-D3).
= 30 ± 1 ns ($\overline{\text{RE}}$).

FIGURE 17. Dynamic functional test (minimum clock pulse high) waveforms for table XIV.

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